



# AI Mandarin Self-Study

## Buddy

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# the problem the problem

**Self-taught language** learners **often fail** because there is no system that **forces them to stay consistent**.

Before this solution, learners had to search for materials on their own, **manually copy** them into notebooks, and often forgot to review lessons (spaced repetition) since there was no binding schedule.

This problem leads to a high rate of burnout along the way. The solution is crucial because consistency and structured exposure are the key factors in successfully mastering a foreign language.





# our solution

A **proactive AI study companion** that eliminates the administrative friction of learning.

Instead of just passively answering questions, the system automatically generates a tailored learning module, emails the formatted document to the user, and directly injects study sessions and review reminders into their personal calendar to enforce discipline.





# AI Agent

Powered by **IBM Langflow** via REST API and accessed through a dedicated custom web interface (developed using **IBM Bob** as an AI coding assistant). The LLM agent acts as a smart orchestrator: it analyzes the user's learning goal, structures the educational content, and autonomously executes a chain of external tools (creating a Google Doc, sending a Gmail, and blocking time in Google Calendar).

Reduces study preparation time from 45 minutes to under 1 minute. It significantly improves learner retention and discipline through automated calendar integration, shifting the user's focus entirely from planning to executing their daily study habits.

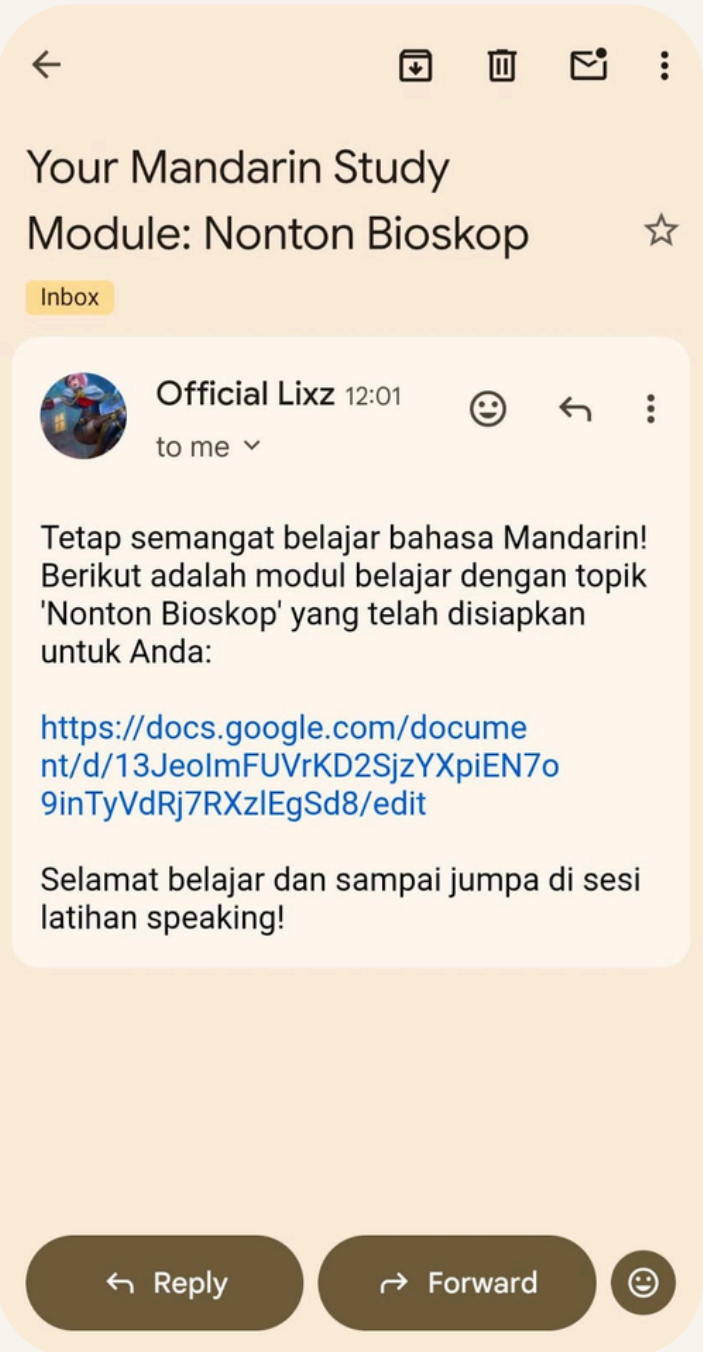
## Impact

# The result

## THE USER INTERFACE



## THE E-MAIL NOTIFICATION



## THE CALENDER REMINDER



# our documentation

<https://github.com/RenceLix/ai-mandarin-buddy>

## RenceLix/**ai-mandarin-buddy**



Antarmuka chat berbasis web untuk belajar bahasa Mandarin menggunakan AI. Frontend statis (HTML/CSS/JS) yang terhubung ke backend **Langflow** melalui REST...

1 Contributor 0 Issues 0 Stars 0 Forks



**RenceLix/ai-mandarin-buddy: Antarmuka chat berbasis web untuk belajar bahasa Mandarin menggunakan AI. Frontend statis (HTML/CSS/JS)...**

Antarmuka chat berbasis web untuk belajar bahasa Mandarin menggunakan AI. Frontend statis (HTML/CSS/JS) yang terhubung ke backend **Langflow** melalui REST API. - RenceLix/ai-mandarin-buddy



# let's build the future together

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